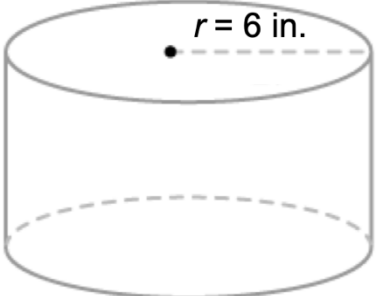
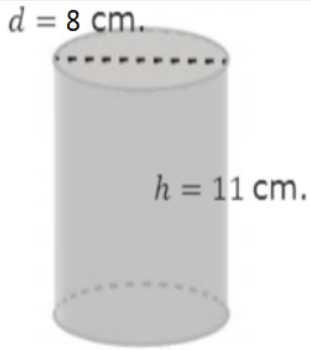


Grade 7
Unit 12
Lesson 3 Practice

Name Answer Key
Date _____
Period _____

1. The formula for surface area of a right circular cylinder is $2\pi rh + 2\pi r^2$. Place a box around the part of the formula that shows the circumference of a circle.

Three right circular cylinders are shown and specific measurements are given. The measurements include either the radius, r , diameter, d , or height, h . For each cylinder, find the surface area in terms of π .

Cylinder	Surface Area
2.  $r = 6 \text{ in.}$ $h = 7 \text{ in.}$	$(2 \cdot \pi \cdot 6^2) + (2 \cdot \pi \cdot 6 \cdot 7)$ $72\pi + 84\pi$ $= 156\pi \text{ in}^2$
3.  $d = 8 \text{ cm.}$ $h = 11 \text{ cm.}$	$(2 \cdot \pi \cdot 4^2) + (2 \cdot \pi \cdot 4 \cdot 11)$ $32\pi + 88\pi$ $= 120\pi \text{ cm}^2$
4.  $h = 1.7 \text{ m.}$ $r = 3 \text{ m.}$	$(2 \cdot \pi \cdot 3^2) + (2 \cdot \pi \cdot 3 \cdot 1.7)$ $18\pi + 10.2\pi$ $= 28.2\pi \text{ m}^2$

Albert wants to send 5 candles to his friends for the holidays. The shipping store explained he must wrap each candle separately in bubble wrap. The bubble wrap is sold in square inches.

5. How many square inches of bubble wrap will he need to purchase for one candle that has a diameter of 6 inches (in.) and is 4.25 in. tall? Use 3.14 for π and round to the nearest hundredth.

$$2(3.14)(3^2) + 2(3.14)(3)(4.25) = 136.59 \text{ in}^2$$

6. How much bubble wrap will he need in for all 5 candles?

$$5(136.59) = 682.95 \text{ in}^2$$

7. Sally bought 2 bongo drums, both look like right circular cylinders. She wants to wrap each one separately in wrapping paper. How much wrapping paper does she need if each drum has a diameter of 7.5 inches (in.) and a height of 10 in.? Use $\frac{22}{7}$ for π and round to the nearest hundredth.

$$2\left(\frac{22}{7}\right)(3.75^2) + 2\left(\frac{22}{7}\right)(3.75)(10) \approx 324.11 \text{ in}^2 \text{ for one drum}$$

$$(324.11)(2) = 648.22 \text{ in}^2 \text{ for both drums}$$

8. An oil tank is the shape of a right circular cylinder and has a diameter of 8.4 yards. The surface area of the oil tank is about 255.85 yd². What is the height of the oil tank? Use 3.14 for π and round to the nearest tenth.

$$255.85 = 2(3.14)(4.2^2) + 2(3.14)(4.2)h$$

$$255.85 = 110.7792 + 26.376h$$

$$h \approx 5.5 \text{ yd}$$

Corey is painting a vase for Mother's Day in the shape of a right circular cylinder. The circumference of the vase is 25.12 in. and has a height of 10.5 in.

9. What is the diameter of the vase?

$$25.12 = 2\left(\frac{22}{7}\right)r$$

$$r \approx 4 \text{ in}$$

$$d = 8 \text{ in}$$

10. How many square inches of the vase does she need to paint not including the top? Use $\frac{22}{7}$ for π and round to the nearest tenth.

$$\left(\frac{22}{7}\right)(4^2) + 2\left(\frac{22}{7}\right)(4)(10.5) \approx 314.29 \text{ in}^2$$